LESSON PLAN · SCRATCH (GR 3–5) ADVANCED: ARCADE ENGINEERS

Lesson 10: My Blocks: Make Your Own Blocks

Module 4 | 45 minutes | Custom blocks (procedures) | Grades 3–5 · ages 9–11

Objective: Define a custom block with an input, call it several times with different values, and explain how it removes repeated code.

Success criteria

- I can define a custom block.
- I can give it an input and use the input.
- I can explain why custom blocks help.

Key vocabulary

- **procedure**: A named group of blocks you can use again (a My Block).
- **define**: The hat that says what your block does.
- **input (parameter)**: A value you give the block when you use it.
- **abstraction**: Hiding the details behind a simple name.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Dance move (5 min): define “clap (times)”; the class performs clap (2), clap (5).
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	The square machine: learners follow the steps.
14 min	**Perform**	Code Lab: square machine, Write once
9 min	**Probe**	Code Lab: greeter block, activity
4 min	**Wrap-up**	Exit ticket: Which hat holds the code of your custom block?

Unplugged warm-up

Dance move (5 min): define “clap (times)”; the class performs clap (2), clap (5).

Teaching the idea

My Blocks are procedures: a named, reusable group of statements with optional parameters. They support abstraction (hide detail), reuse and single-point fixes. The define hat holds the body; inputs are reporters scoped to that script.

Analogy: A recipe card: “make pancakes (number)” — you write the steps once and cook any amount.

Live demo: Draw three squares the long way (12 move/turn blocks), then refactor into a square block. Count the blocks before and after.

Questions to ask

Ask	Listen for
Where do you put the code your block runs?	Under the define hat.
What is the input for?	To change what the block does each time.
Why are My Blocks useful?	Shorter code, reuse and one place to fix bugs.

Differentiation

Support
Provide the define hat; learners fill in the body and use it.

Stretch
Make polygon (sides, size) with two ideas: turn 360 / sides.

Common misconceptions

Misconception	How to fix it
Defining a block runs it	It only runs when the block is used.
The input is a variable for the whole program	It belongs to that define script.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Which hat holds the code of your custom block?

Homework
Write the steps of a custom block for something at home, like “brush teeth (minutes)”.

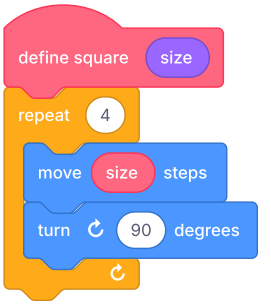
Answer key — Kitabash activities

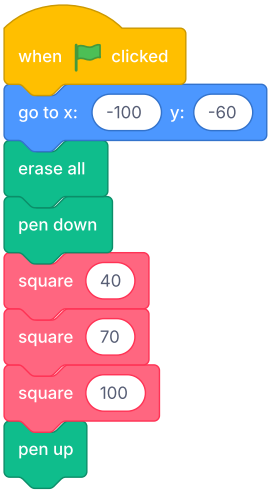
1. Code Lab: square machine (perform · code lab)

Goals checked: Defines and uses a block · Uses the input · Draws 3 squares

One solution:

Pencil





2. Write once (perform · multiple choice)

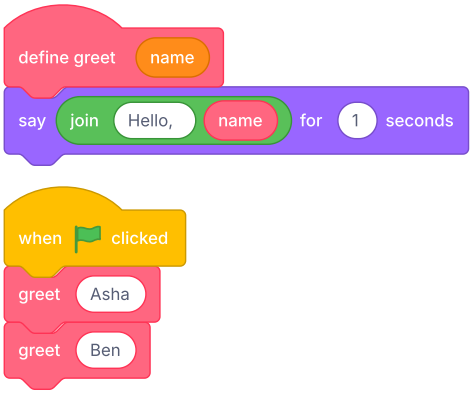
Answer: Once

3. Code Lab: greeter block (probe · code lab)

Goals checked: Defines and uses greet · Greets Asha, then Ben · Uses the input

One solution:

Robot



4. Explain your thinking (probe · free response)

Look for: block, input, define. **Example:** In my platformer I would make a block called jump with an input called height. I could use jump (10) for a small hop and jump (20) for a big jump, and if the jump has a bug I only fix the define script.

Answer key — worksheet

1. **Key words:** procedure = A named group of blocks you can use again (a My Block). · define = The hat that says what your block does. · input (parameter) = A value you give the block when you use it. · abstraction = Hiding the details behind a simple name.
2. **Plan your code:** Goals: Defines and uses a block; Uses the input; Draws 3 squares.
3. **Write once:** A. Once — It's written once in the define script and runs each time the block is used.
4. **Plan your code:** Goals: Defines and uses greet; Greets Asha, then Ben; Uses the input.
5. **Explain your thinking:** In my platformer I would make a block called jump with an input called height. I could use jump (10) for a small hop and jump (20) for a big jump, and if the jump has a bug I only fix the define script.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.